



Printed Electric Twin-Fuselage Research Aircraft

IMPERIAL COLLEGE LONDON

DEPARTMENT OF AERONAUTICS

GROUP DESIGN PROJECT

3D Printed UAV Testbed

Design and validation of the wing structure

FATT04 - Structures



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Abstract

This report details the design and analysis of a wing structure for an additively manufactured UAV testbed. Aerodynamic and inertial loads are determined analytically and concurrent tip deflections are calculated and shown to be in good agreement with known approximations. Finite element analysis and topology optimisation procedures are outlined, and outcomes from topology optimisation are shown to be insufficient to meet the requirements from initial sizing of the aircraft. Spars, skin and torque-bearing 3D-printed parts are sized and a final analysis of the wing structure gives a maximum tip deflection of 79.3 mm and maximum stress of 25.7 MPa with a total mass of 5.77 kg, meeting the objectives set out at the start of the project.

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1 Introduction

1.1 Overview of the project

The aim of this project was to develop an unmanned aerial vehicle (UAV) which would act as a testbed for several areas of research within the Department of Aeronautics at Imperial College London. The UAV was to be made using additive manufacturing and had to satisfy the drone regulations of the Civil Aviation Authority (CAA). From a 'Structures' perspective, it was found from conversations with academics that topics such as morphing wings [1] and structural health monitoring [2] were incorporable on the aircraft as the test equipment does not necessarily require excessive mass. Additionally, there is scope for 'proof-of-concept'-type tests, where a researcher may wish to show that their work can function in flight. To this end, the mission profile consists of a 30-minute cruise segment at 35 m/s. Initial sizing was performed by Functional Area Technical Team (FATT) 01 [3] and the key parameters are shown in Table 1.

Table 1: Key parameters from initial sizing.

Wing loading / Nm^{-2}	345	MTOW / kg	78.12
Power-to-weight / WN^{-1}	6	Payload mass / kg	7.39
Wing ref. area / m^2	2.02	Aspect ratio	10
Cruise speed / ms^{-1}	35	Wingspan / m	4.49

Additionally, it should be noted that this aircraft is not conventional. Several hours of team-wide discussion resulted in the design shown in Figure 1, which features a central test-section (shown in green) attached to the fuselage via two spars, two fuselages (which house batteries and systems such as the camera and flight controller), wing-mounted propellers and an 'n'-shaped empennage.



Figure 1: Render of the final configuration.

Some attempts have been made to design and manufacture 3D-printed autonomous aircraft. Examples in education include the University of Southampton's 'SULSA' aircraft [4] (which utilised laser sintering), the University of Virginia's 'Razor' projects [5] (a series of blended-wing body designs) and the University of Sheffield's nine-part, 1.5-wingspan blended-wing endeavour [6]. Industrial examples are harder to find given that UAVs have largely military applications, but exceptions such as Aurora Flight Sciences' 150mph-top-speed 3D-printed drone [7] show the capabilities of additively manufactured UAVs. The key difference between these UAVs and the one discussed in this report is size: with a 4.49 m wingspan, the UAV considered in this report is considerably larger, and hence poses a different set of challenges. Using the CAA's drone regulations a flight envelope was developed by Lau [8], giving a range of load factors n between 4.26 and -2.26. The magnitude of the load factors is reasonably high and hence inertial loads from the wing's self-weight play a very important role in the design process.

1.2 Additive manufacturing overview

Additive manufacturing (AM) is a fairly new development and has found increased use in recent years thanks to its versatility, since parts manufactured using AM do not adhere to the same constraints as traditional manufacturing methods. AM may also enable a rapid prototyping approach as it is particularly economical when manufacturing a small number of more complex parts. AM does, however, have downsides. One of the key

constraints for 3D-printed parts is the print size. To minimise the number of parts required to be printed and hence make assembly simpler, the design was developed under the assumption that parts could be printed using the Department of Aeronautics' Stratasys Fortus 450mc, which has a large print volume of 406 x 355 x 406 mm.

The additive manufacturing method of the most interest in the case of this project was Fused Deposition Modelling (FDM) due to its widespread use. FDM is an extrusion-based process where a thermoplastic filament is melted and deposited by the printer in a pre-defined path, layer by layer. One of the key weaknesses of FDM is its susceptibility to anisotropy, meaning the parts are considerably weaker in the printer's z-direction (the direction in which it builds layers) [9] - this increases the complexity of accurately modelling structural 3D-printed parts. Additional complications such as print angles and overhang must also be taken into consideration: an overhanging feature with an angle of less than 45 degrees to the print plane would require support structure, increasing print time and cost without any structural benefits. Therefore, the amount of support structure should be minimised where possible. Finally, 'infill' can be used to make parts faster and cheaper. This entails printing a given percentage of the solid material within the same bounds, and can be varied to give the desired structural properties for minimum cost.

1.3 Information on topology optimisation

Topology optimisation is another relatively recent innovation in structural design which aims to minimise (or maximise) an objective function subject to user-defined constraints by removing material from the source part. A common approach is to minimise compliance while defining a volume constraint that may not be exceeded, giving the stiffest part which meets the required mass. Additionally, geometric constraints may be defined so that the resultant part adheres to manufacturing constraints. A more in-depth discussion of the mathematical foundations of topology optimisation is given by Sigmund and Maute [10]. Frequently, the approach utilised is 'Solid Isotropic Material with Penalisation' (SIMP), whereby each finite element within the problem is assigned a value between 0 (representing a 'void' - material which can be omitted) and 1 (representing a 'solid' - material which should be maintained) [11].

There have been successful attempts to use topology optimisation in conjunction with the structural design of small aircraft, as shown by Taylor et al [12]. Topology optimisation generally does not, however, incorporate considerations for buckling (which are computationally expensive). Eves et al [13] have sought to solve this issue by using a combined load case approach wherein a second, pressure-based case is introduced with a deformation constraint to prevent buckling.

1.4 Objectives

The objectives of the Structures sub-team were to produce a wing structure:

- which could endure all conditions within the flight envelope without exceeding critical stresses or a 100 mm tip deflection.
- which had a mass of 6 kg or less.
- which played to the strengths of additive manufacturing.

2 Methodology

2.1 Layout

Given that the wing dimensions provided by the Aerodynamics sub-team had a root chord of 0.69 m and a half-span of 2.25 m, 3D printing would require the wing to be manufactured in several parts and assembled afterwards. Additionally, the wing features high lift devices and ailerons which must be actuated. The layout is shown below in Figure 2, and was determined by enforcing breaks at hinge locations and the propeller mounting rib, and then slicing the parts in between so that their maximum dimension would not exceed that of the Fortus

450mc. It should be noted that ailerons and high lift devices take up 30% of the chord, and that there are 5 mm ribs between adjacent sections (except for around **2(M)**) to mount the hinges for trailing edge device actuation.

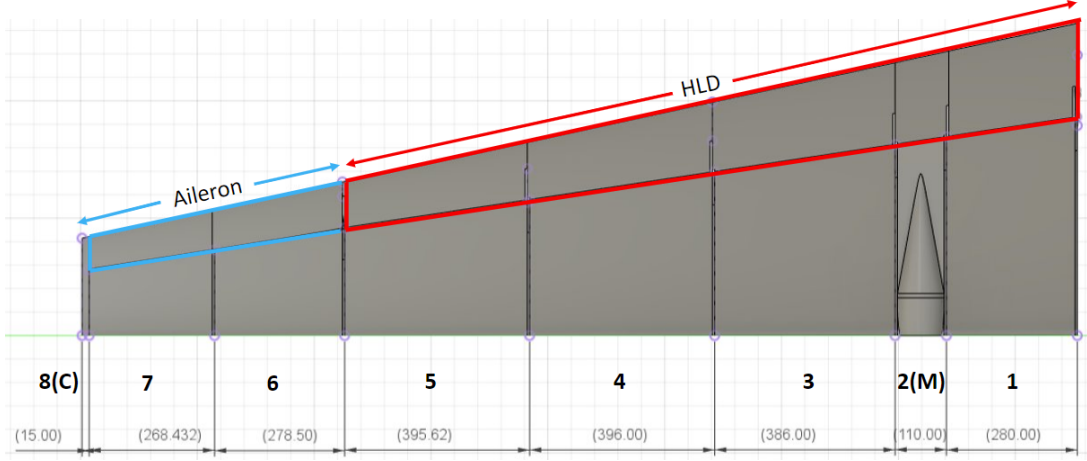


Figure 2: Separation of the wing into printable parts [14].

2.2 Analysing loads

A diagram of the loads is shown in Figure 3, depicting inertial loads from the wing weight and propeller (of mass m_p), and lift. It should be observed that $()'$ denotes a sectional property. Several iterations of the aircraft were designed and hence the planform and lift distribution changed frequently throughout the project. For analytical purposes, an elliptical lift distribution was assumed before using results obtained from XFLR5 [15], an aerofoil analysis tool, later in the design process - a quadratic mass distribution was also assumed. To optimise the wing it was necessary to have an accurate analysis of the loads acting upon it. To calculate the shear force S and bending moment M , the following equations were used:

$$S_i = \int_0^{y_i} L'(y) - n g m'(y) dy \quad (1)$$

$$M_i = \int_0^{y_i} S(y) dy \quad (2)$$

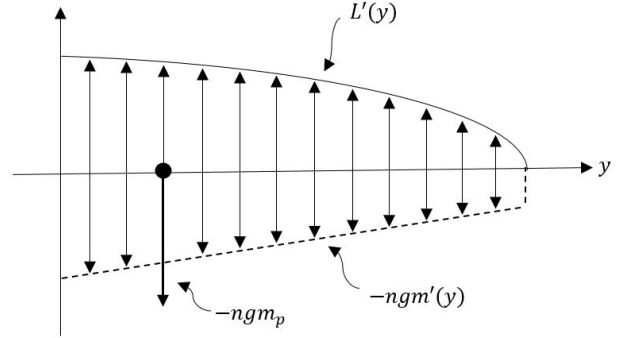


Figure 3: Depiction of loads on the wing.

while the torque T distribution was found by integrating the contributions from lift, inertia and the aerofoil's self-moment along the span. Deflections across the span were calculated analytically using Euler-Bernoulli Beam theory

$$M = -EI \frac{d^2 w}{d^2 y} \quad (3)$$

where M is the bending moment, E is the material's Young's modulus, I is the wing section's second moment of area, w is the deflection and y is the spanwise coordinate. This has some inaccuracies: the value of E was assumed to be the same as the isotropic material (which is optimistic given the anisotropic nature of fused-deposition modelling) and the value of I along the span was taken to be that of the local aerofoil section - with cutouts for weight saving, this would be reduced. Additionally, approximations from the Massachusetts Institute of Technology [16] were used which made assumptions such as the net aerodynamic loading $L'(y) - n g m'(y)$ being proportional to the local chord length $c(y)$ and neglected the point mass of the propeller. However, use of these methods is simple and allows for verification of the more complex (and hopefully accurate) calculation. The formulae give tip deflections δ_{tip} as below:

$$\delta_{tip} \approx \frac{M_0}{EI_0} \frac{b^2}{8} \quad (4)$$

$$\delta_{tip} \approx \frac{L - nm_{wing}}{E} \frac{1}{K_I c_{avg}^4} \frac{1}{\tau(\tau^2 + \epsilon^2)} \frac{b^3}{96} \frac{1 + 2\lambda}{1 + \lambda} \quad (5)$$

where $(\cdot)_0$ denotes values at the wing root, b is the wingspan, K_I is approximated as 0.036 and relates the aerofoil's thickness and chord to the second moment of area, τ and ϵ are the thickness and camber ratios, and λ is the taper ratio.

2.3 Lifting surface design theory

Some analytical formulae were also used to either influence or validate the design. Namely, the dimensions of through-thickness elements were determined using shear flow analysis derived from Iannucci [17]. This involved combining shear force and torque contributions about the aerofoil's shear centre to determine the shear flow q of 3D-printed 'pseudo-spars' at the 25% and 70% chord locations along the span.

$$q = \frac{S}{2h_w} \pm \frac{T}{2h_w w_w} \quad (6)$$

Here, h_w and w_w denote the wingbox height and width (chordwise) respectively. Torque contributes positively to the front spar and negatively to the rear spar, and the required thickness may be calculated using Equation (7), as found from ESDU 71005 [18].

$$t^3 = \frac{qh_w}{8.1E} \quad (7)$$

Additionally, the stresses on the ribs due to crushing loads were evaluated using Equation (8)

$$\sigma = \frac{M^2 s h_c t_e}{2EI^2 t_r} \quad (8)$$

where s is the distance between adjacent ribs, h_w is the wingbox height, t_e is the skin thickness and t_r is the rib thickness. These stresses were then compared to the yield and buckling stresses.

2.4 Finite element simulations

A number of simulations were run using Abaqus CAE, an industry standard Finite Element Analysis (FEA) software. Initial simulations used a solid element approach with quadratic tetrahedral elements, so that models which had demonstrated mesh convergence and good agreement with analytical results could then be used in topology optimisation. Using the analysis of the aerodynamic and inertial loads, application in FEA could be performed in several manners, with varying degrees of accuracy. For example, 'Gravity' loads were initially used to model inertial forces from the wing's self-weight, 'Pressure' loads modelled the aerodynamic pressures which were imported from XFLR5 and the 'Concentrated force' load was used to model the inertial load from the propeller. Representative forces and torques at the tip were also found which gave the similar deflections as for the complex case.

Later simulations used shell elements to model the wing by importing aerofoil coordinates to sketches via a script developed in Python, and then lofting between the sections to create the skin. Further lofts and planar shells were then used to create the spar and rib structures respectively. Sections could then be assigned to each set to give the correct material properties and thicknesses. For these simulations quadrilateral mesh elements were used with a sweep in the spanwise direction to account for the taper. Once again, multiple methods were tested for applying loads and mesh convergence studies were performed on all simulations.

2.5 Topology optimisation procedure

Abaqus CAE was also used to perform topology optimisation on the wing structure. For this procedure, an initial STEP file of the wing planform (without trailing edge devices) was used which was entirely solid. From this part, material could be removed to meet the mass target of 6 kg. As stated previously, tetrahedral elements were used on a solid model of the wing. The wing was optimised for the most severe load case from the flight envelope, which was found to be the -2.26 load factor scenario. The concurrent loads were applied in different ways by using combinations of the load types detailed above. A SIMP-based approach was used on a compliance problem, which meant setting an objective function to minimise strain energy. A volume fraction constraint was determined by taking the target mass and dividing by the actual mass of the solid wing, giving a value of 0.214. Additionally, some geometric constraints were used, such as an envelope for the member size to ensure that parts were of manufacturable dimensions, and a stress constraint at the yield strength of ASA. Void material was assigned strength properties a factor of 10^6 lower than those of the solid material, and the number of design cycles was most often set to around 20, which was a good medium between convergence and computational time. Results from the topology optimisation for stresses, strains and displacements were observed, and an STL file was extracted. Using Fusion 360 CAD software, the design was adapted so that it was manufacturable by 3D printing. This design was then reanalysed using Abaqus and the results were compared.

3 Results and Discussion

It should be noted by the reader that material selection for the wing was performed by Fong [14], and that Acrylonitrile Styrene Acrylate (ASA), a conventional thermoplastic used for 3D printing, was to be used for parts made with additive manufacturing. Some of the material properties of ASA are shown in Table 2. For modelling purposes, a factor of 0.9 was applied to the Young's Modulus E in order to account for anisotropy.

Table 2: ASA material properties.

Density ρ / kgm^{-3}	E / GPa	Yield Stress / MPa
1070	2.00	28.3

3.1 Load approximation and analysis

Loads were calculated for both the 4.26 and -2.26 load factors and it was found that the -2.26 case was more limiting since the inertial loads act in the same direction as the lifting loads. Initially, an elliptical lift distribution was used to calculate deflections, however when validating the design a more accurate distribution from XFLR5 was incorporated. Both of these lifting loads, as well as the inertial forces, are shown in Figure 4.

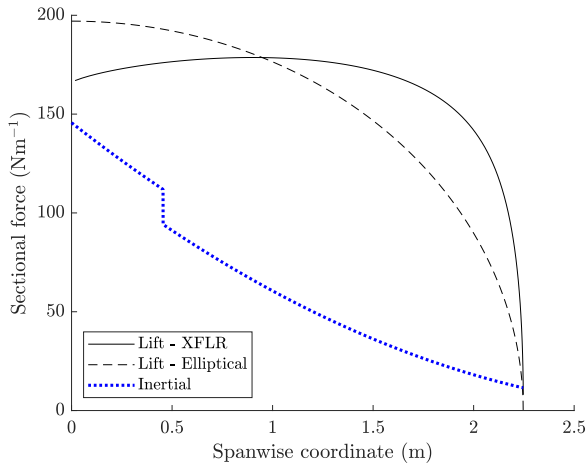


Figure 4: Loads over the span of the wing.

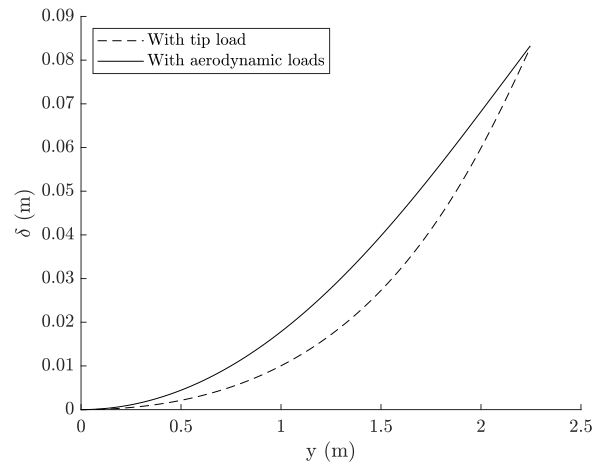


Figure 5: Deflections given by actual and tip loads.

Using Equations (1)-(3) it was possible to calculate the shear force, bending moment and torque distributions across the span, as well as the deflections δ . All of these loads are for a solid wing, which is inaccurate but

important to analyse so that the deflections given by the pre-optimisation Abaqus model could be verified. In order to simplify the applications of loads in Abaqus, it was beneficial to approximate the deflections via a single point load and point torque which could be applied to the tip of the wing. Representative loads and torques of 124 N and 67.5 Nm were calculated respectively, and a comparison of the deflections from these loads to the actual loads is shown in Figure 5. As expected, the mid-span deflections are lower due to the absence of the distributed upwards load when approximated with forces at the tip.

Tip deflections δ were compared to the approximations presented in Equations (4) and (5) and the results are shown in Table 3. The maximum error between results is 3.7%, showing good agreement and demonstrating that the analytical modelling of the wing loads is accurate.

Table 3: Analytical and approximate tip deflections.

δ_{tip} - analytical (mm)	δ_{tip} - Equation (4) (mm)	δ_{tip} - Equation (5) (mm)
82.9	86.0	83.1

3.2 FEA: Validation of mesh and load comparison

Ten simulations were run using the initial solid model before use in the topology optimisation, with the number of elements ranging from 2817 to 112387 (the maximum before exceeding the node limit of the student license). The variation of the resulting tip deflections is shown in Figure 6 and demonstrates that the simulations do converge to give a value of 87.75 ± 0.01 mm. This represents a 5.5% difference from the analytical value, which may be due to the quadratic mass assumption made during load analysis not being replicated in the FE model.

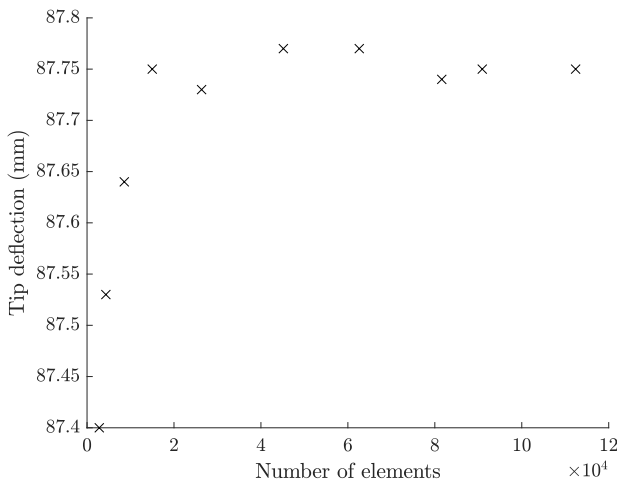


Figure 6: Tip deflection against number of elements.

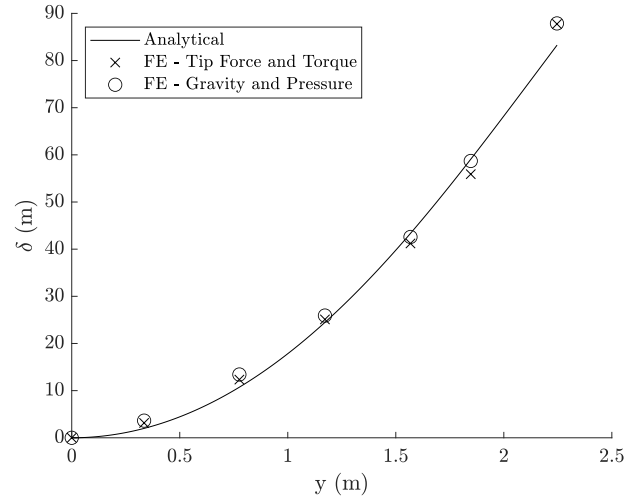


Figure 7: Analytical and FEA deflections.

A comparison between the deflections calculated analytically, those from Abaqus with a tip force and torque, and those from Abaqus with 'Gravity' and 'Pressure' loads, is shown in Figure 7. The points shown for the Finite Element simulations correspond to the rib locations. Both sets of results from FEA display greater deflections near the tip but deviation between the results is small, with a mean percentage difference of 5.44% between the different load cases. As expected, applying distributed loads gave higher deflections at the wing's mid-span. From these results it was determined that the tip load and torque were a good representation of the distributed load and could be used for topology optimisation problems.

3.3 Topology optimisation: problems and iterations

It was desired to formulate the TO problem with very accurate modelling in order to obtain a design which would be optimised for the exact worst-case condition that had been calculated. To this end, 'Pressure' and 'Gravity' loads were used in early TO problems to model the aerodynamic and inertial forces. However, results not satisfactory, as only the surface was preserved. This was due to the Abaqus CAE student license only allowing simulations with less than 250,000 nodes and hence placing a severe limit on the fidelity of the model.

Therefore, by preserving only the surface elements, the volume constraint was nearly exceeded, leaving very little room for other important elements to be identified. These early iterations resulted in very excessive tip deflections and were hence of little use. To avoid the preservation of these surface elements, which could be sized using buckling simulations later in the design process, the tip load and torque method was used by applying the 124 N force and 67.5 Nm torque previously discussed at the shear centre of the wing tip.

Additionally, areas where the ribs would be located were defined as 'frozen' regions from which material could not be removed. Mesh fidelity was initially unnecessarily high at the trailing edge and insufficient nearer the shear centre, where the stress gradients were much greater. The mesh was adjusted accordingly by partitioning at the shear centre and assigning seeds to give a more refined mesh in this area. The tip load and torque were tested individually to ensure that the results were intuitive - tip loads produced a result very closely resembling an I-beam (which are known to perform well in bending) and the torsional case produced many diagonal spar/rib elements. Following this, a final optimisation was run with the target volume constraint of 0.214, a minimum member size of 0.4 mm (so that no extremely thin, hard-to-print parts were produced) and an objective of minimising strain energy. Convergence of the resulting volume fractions and strain energy values is shown in Figure 8.

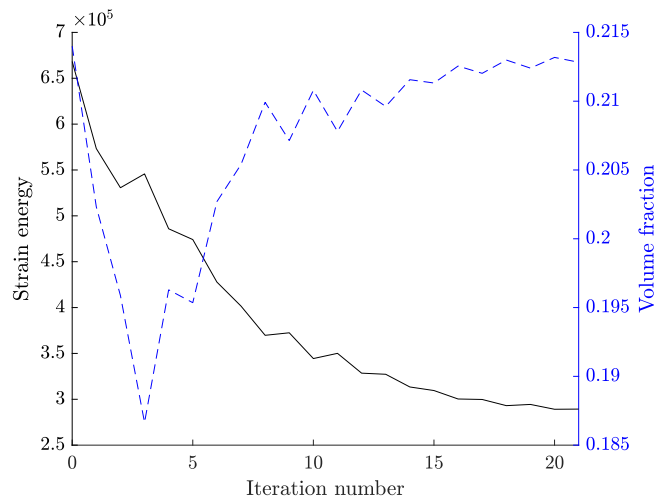


Figure 8: Strain energy and volume fraction over TO iterations.

The final design given by topology optimisation can be seen in Figure 9 - the result was a single spar which was slightly cranked towards the root and had some cutouts. Additionally, the tip separates into thicker parts at the leading and trailing edge to counteract the torsional load applied at the tip. In terms of cross-section, the part develops from an I-beam-like shape at the root to a more rectangular section before it splits off near the tip. This design was modified in Fusion 360 by adding material where 3D printing procedures would require it and by smoothing some of the large sharp edges, meaning the initial part mass of 5.997 kg was increased to 6.343 kg.

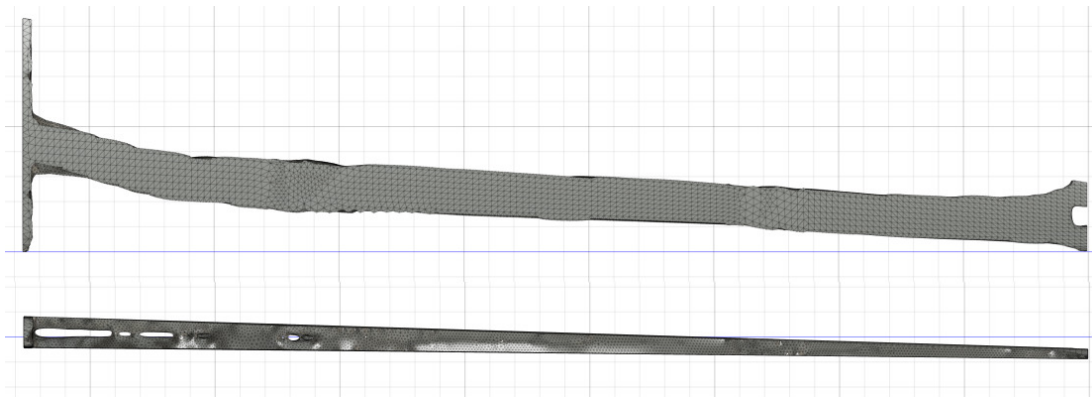


Figure 9: Top and side view of final topology optimisation iteration.

This design gave a tip deflection of over 400 mm and was hence incapable of meeting the deflection requirements set out at the start of the project. A further optimisation was run using Nylon 12 CF, a carbon fibre-reinforced Fortus 450mc printing material with better strength properties to try and obtain a printable structural part.

The tip deflections and masses are shown in Table 4 alongside the values for an 18 mm carbon fibre reinforced polymer (CFRP) rod. It can easily be observed that the CFRP rod is far more mass-efficient than the 3D-printed parts and hence there was evidence that perhaps 3D printing the main structural part of the wing was not the best solution for the aircraft being designed.

Table 4: Comparison of tip deflections and masses for different structural components.

	TO: ASA	TO: Nylon 12 CF	CFRP Spar (18mm diameter)
δ_{tip} (mm)	420.4	87.5	98.5
Mass (kg)	6.343	6.284	0.618

3.4 Reducing mass and sizing structural components

CFRP spars are commonly used in UAVs as they perform very well in bending due to their high Young's Moduli. Given the discrepancy between the E values of CFRP and 3D-printed materials, CFRP parts can produce similar performance with far less volume, and hence mass, required. Instead of sizing a spar which would be tapered for the most mass-efficient design, it was decided that commercially available products be used as they are very cost-efficient and could still fulfil the structural requirements of the UAV being designed. The limiting factors with regards to spar size were thus:

- the commercial availability of required size.
- the thickness of the aerofoil at 25% chord along the span.

The thickness of the aerofoil section is a limitation on spar size since a spar cannot exceed that dimension. With this in mind, the largest constant-section spar which could be used was a circular-section 18mm CFRP rod. This is what was eventually opted for, as it provided maximum stiffness and was easily sourced.

The purpose of the CFRP spar is to resist bending loads and it is placed at the 25% chord location, which lies aft of the the range offered by Niu [19], but since the layout is effectively single spar for simplicity and mass-saving, it makes sense to shift this location back slightly. A diagram of the internal wing structure is shown in Figure 10, with 3D-printed parts shown in red and the CFRP spar in blue. As can be seen in Figure 10, a 3D-printed



Figure 10: Sectional layout of the wing.

'pseudo-spar' is located at the 70% chord location - the termination of the main wingbox, since trailing edge devices take up the other 30% of the chord. This spar, along with the material surrounding the front spar, must be sized in order to be able to resist the torsional loads using Equations (6) and (7). The required thickness distributions for the pseudo-spars are shown in Figure 11, showing decreasing requirements towards the tip, as expected. The discontinuities on the rear spar thickness are caused by the torque and shear force contributions to the shear flow cancelling each other, and the Fortus 450mc print limit of 0.127 mm is also shown. It would be preferred to not print very thin parts or increase thickness when printing in the z-direction and given the low required thicknesses, tapering the pseudo-spars would not save much mass. Therefore, constant thicknesses of 3 mm and 2 mm were chosen to be used for the front and rear pseudo-spars respectively.

The skin thickness must also be determined so that it does not buckle and the aerofoil shape is maintained. This was done by simulating the loads on the wing in Abaqus increasing the skin thickness in steps of either 0.127 mm or 0.178 mm to replicate the dimensions that could be performed by the Fortus 450mc. Then, the eigenvalues were analysed and the first simulation to have a lowest eigenvalue greater than 1 (meaning that the loads would have to be increased to cause buckling) would be the lowest possible skin thickness to not buckle, giving the lowest mass - results are shown in Figure 12.

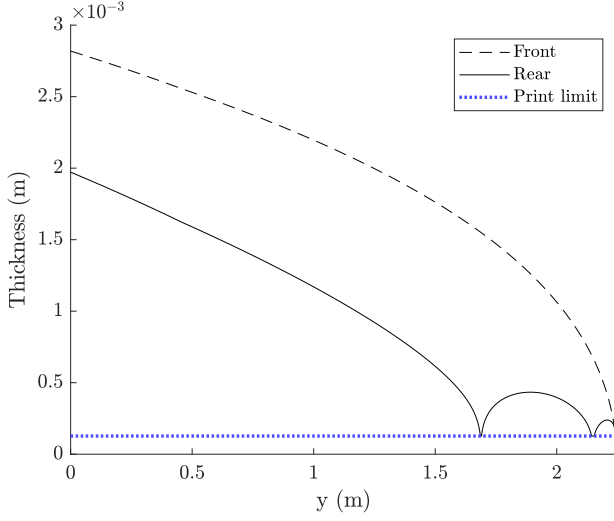


Figure 11: Pseudo-spar required thicknesses.

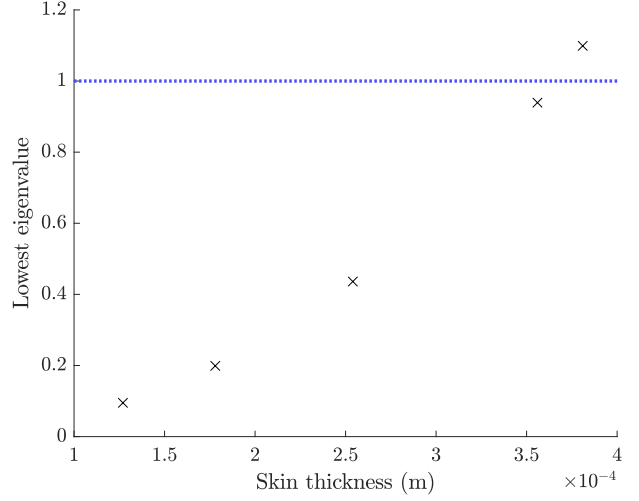


Figure 12: Lowest eigenvalues for skin thicknesses.

A skin thickness of 0.381 mm was chosen, which gave a lowest eigenvalue of 1.0989 corresponding to the buckling of skin near the root. All of the first 100 eigenvalues were examples of skin plate buckling between the ribs, and no primary buckling modes were found within this range, meaning the eigenvalue for any primary buckling mode exceeded 1.89. This is desirable as primary buckling is a critical point of failure for the wing, unlike the secondary buckling of the skin which may have an earlier onset. Given the low required skin thickness of 0.381 mm, it was determined that the use of stringers to lower the mass was not necessary (especially as these parts would be very thin, and hence not particularly suitable for printing in the z-direction).

Rearranging Equation (8), it was possible to find required rib thicknesses given the bending moment and second moments of area at the pre-defined rib sections. In all cases, the required rib thicknesses were of the order of micrometres, and hence the 5 mm thickness set by the requirement to attach hinges means the ribs are well over-designed for crushing loads. For a lower mass design, fewer ribs could be used if a different hinge mechanism allowed.

3.5 Final design performance

The final design hence features:

- A 2 m solid circular-section CFRP spar of 18 mm diameter to resist bending and some torsion.
- A 3 mm thickness ASA pseudo-spar surrounding the main CFRP spar, resisting torsion and shear forces.
- A 2 mm thickness ASA rear pseudo-spar to resist torsion.
- 0.381 mm thickness ASA skin to maintain aerofoil shape.
- 5 mm thickness ASA ribs at the $y = 776, 1172, 1568$ and 1847 mm locations to take crushing loads, mount hinges for trailing edge devices and transfer loads to the main spar.
- An 80 mm thickness ASA rib to mount the propeller assembly.
- An ASA rib of 15 mm thickness at the tip of the wing.

The wing is modular in the sense that it is printed in the parts outlined earlier in this report, and dovetail joints are used to align the adjacent parts while epoxy sets and provides a very strong bond - these joints were designed by Fong [14].

Deflections for both the -2.26 and 4.26 load factors are shown in Figure 13, giving a marked performance increase from the design given by topology optimisation, with a final tip deflection of 79.3 mm. Results in Table 5 demonstrate that the structure designed fulfils all numerical objectives determined at the outset of the project.

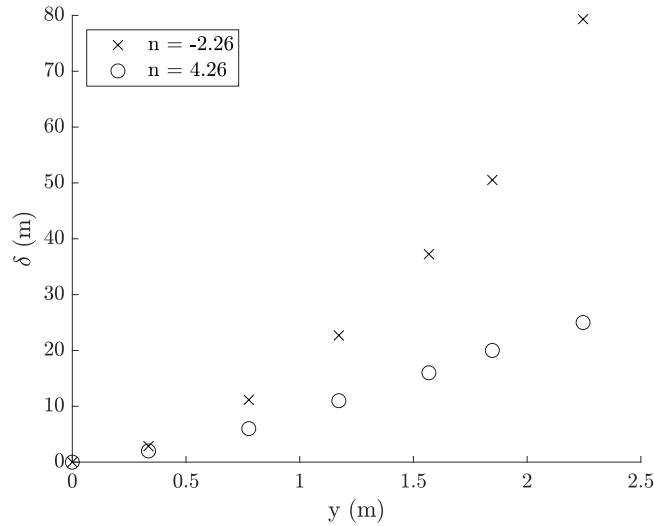


Figure 13: Deflections of final design at rib locations.

Table 5: Comparison of the final results and objectives.

Maximum deflection (mm)		Maximum stress (MPa)		Mass (kg)	
Actual	Target	Actual	Target	Actual	Target
79.3	100.0	25.7	28.3	5.77	6.00

3.6 Further design considerations

Given the time constraints associated with this project, there is much scope for improvement of the design. For example, a detailed design of the D-section was not considered. Although the D-section has the same thickness as the rest of the skin and hence should not buckle due to its curvature making it less prone to buckling [20], it is also desired that the D-section has some resistance to damage. This is so that, in the event of collisions, the failure is not catastrophic. It was proposed that tubulanes could be used in the D-section as they are known to be very damage-resistant and manufacturable with 3D printers [21].

Additionally, the use of lattice structures was proposed due to the potential to create high-specific stiffness designs [22]. These structures are particularly well-suited to additive manufacturing due to their complexity. Additionally, the lattice structure geometry can be adhered to meet given requirements - for example, it may be desired to increase the volume fraction near the spar locations for bending and torsional loads, and have a thinner, sparser lattice in between where less material is required and mass can be saved. Furthermore, they could be used in a stringer-like fashion in order to decrease the skin thickness by increasing the effective thickness of the skin. The capacity of lattice structures to be multifunctional in resisting bending, torsional and compressive loads while being lightweight makes them very appealing. Due to the inherent complexity of modelling lattice structures, which is an active area of research [23], the analysis and optimisation of lattice structures was not performed for this design but should be considered in future design iterations.

Prototyping is an important phase in the additive manufacturing process, especially in this case due to the inaccuracy of the modelling in design. Some attempts were made during the later stages of this Group Design Project to test parts of the wing. The wing tip was printed on the Department of Aeronautics' Ultimaker 3 Extended 3D printer, a more economically viable option than the Fortus 450mc for prototyping, which had a smaller print bed of 215 x 215 x 200 mm. The printed part featured wire channels, a dovetail joint for connection, and a variable thickness Hexa Grid lattice structure design using Autodesk Netfabb, a software for pre-processing parts for additive manufacturing. Netfabb does not have many of the more common lattice unit cells and does not have the capability of analysing lattice structures - hence, the printed lattice structure is merely demonstrative. The resulting part is very stiff and lightweight, and is shown in Figure 14. It was printed with minimum layer height to give the best surface finish, as well as a reduced nozzle speed, but some post-processing would be recommended in order to smooth the surface, which is visibly quite rough in the spanwise direction. A rib was also printed to test the joining of two parts, and the parts mated successfully, although this required some filing of the dovetail joint. The resulting interface was very strong and it is recommended that testing could be performed to determine whether epoxy, and its pertaining mass, is necessary.

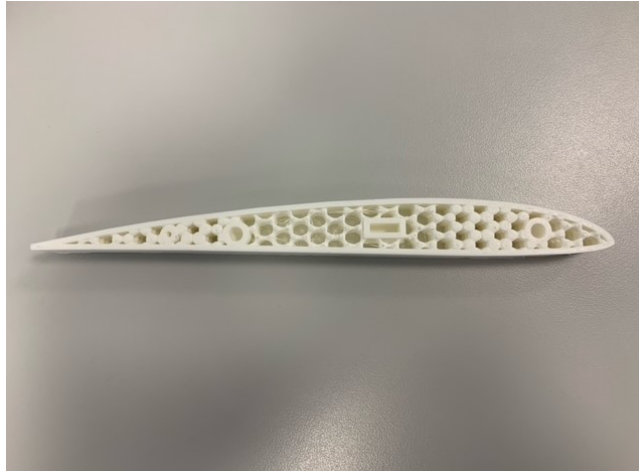


Figure 14: 3D printed wing tip with variable thickness lattice structure.

4 Concluding remarks

Both analytical and computational methods were used to size the key structural components for a 4.49 m wingspan UAV. Approximations of the loads were compared to the analytical results and were found to be within a reasonable range of each other and Finite Element analyses showed further agreement with the analytical results, demonstrating a good correlation and also that approximation of the actual loads on the wing by a torque and force at the tip was viable for design and analysis. Topology optimisation was used with these approximate loads to design the 3D-printed structure of the wing, but this was found to not meet the requirements defined by FATT 01. Instead, a CFRP spar was used along with parts printed using ASA: a skin sized to resist buckling, pseudo-spars to resist torsional loads, and ribs sized for hinge attachment. FEA simulations were then run on this final design, showing that all criteria were met, with a final mass of 5.77 kg.

There were some inaccuracies in the modelling of the wing structure throughout this project. Namely, the anisotropy of 3D printed materials was accounted for simply by applying a factor of 0.9 to all material strength properties. Although this gives a reasonable safety factor, it is a fairly crude approximation which could be more accurate. That being said, the level of anisotropy varies due to material and printing settings such as nozzle speed, so accurate modelling is very difficult. Additionally, approximate loads were used which, although shown to be in good agreement with the more accurate loads from XFLR5, were also found to underestimate the mid-span deflections. Use of a distributed load would have potentially provided a better approximation of the actual load case. Buckling was not considered in the topology optimisation, although this did not affect the result since the final design iteration given by Abaqus did not feature any slender members in compression. The final constraint was the number of nodes in Abaqus, which affected results given by topology optimisation simulations but not FEA simulations, which were all seen to converge. Because only 250,000 nodes were available, the level of intricacy on the wing, which was large and had reasonably complex geometry, was fairly low.

Given more time, the design of the wing could be far more intricate and suited specifically to the load requirements given by the Aerodynamics sub-team. An ideal topology optimisation setup would be to have an optimisation problem with shell elements for the preserved wing surface, to which the pressure loads are applied, and solid elements for the internal structure of the wing, from which material may be removed. This would allow for an optimisation based on more accurate loading and, if performed with more tetrahedral elements, could give better performance. However, it is likely that the material properties of 3D printable material simply do not satisfy the specific stiffness constraints inherent to the project. A more accurate method for modelling the anisotropy would also be preferable, perhaps by testing parts printed using similar settings to the final wing and adjusting material properties accordingly. This is still somewhat imprecise, but would be an improvement on the current method. The incorporation of lattice structures into topology optimisation could provide a design with higher specific stiffness, but this would also increase the complexity and be computationally expensive.

The resulting design nevertheless met the criteria defined at the outset of the project, but much time was spent on designs which were ultimately not feasible due to the poorer strength properties of additive manufacturing materials when compared to commonly used materials such as carbon fibre reinforced polymers. The size of the aircraft, and the loads encountered, perhaps do not lend themselves to an entirely 3D-printed structural design, but 3D printing can still be useful when it comes to rapid manufacturing and ease of assembly.

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A Flight envelope

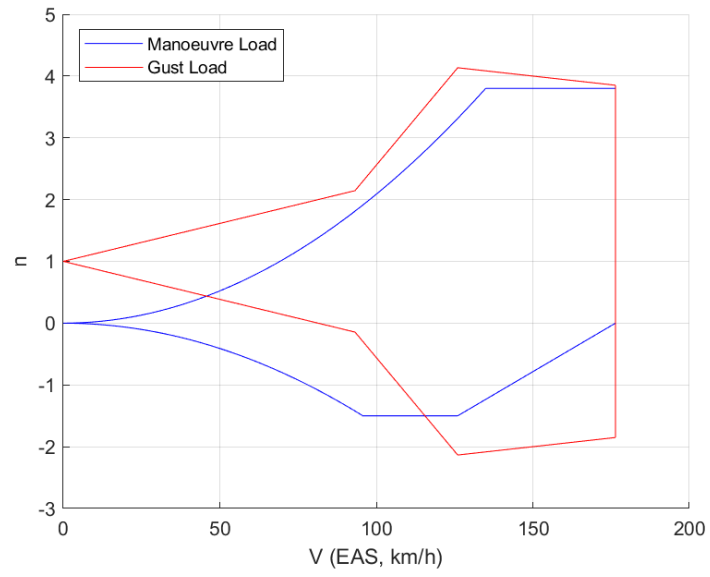


Figure 15: Flight envelope with a load factor range of -2.26 to 4.26 [8].

B Shear Force, Bending Moment and Torque Distributions

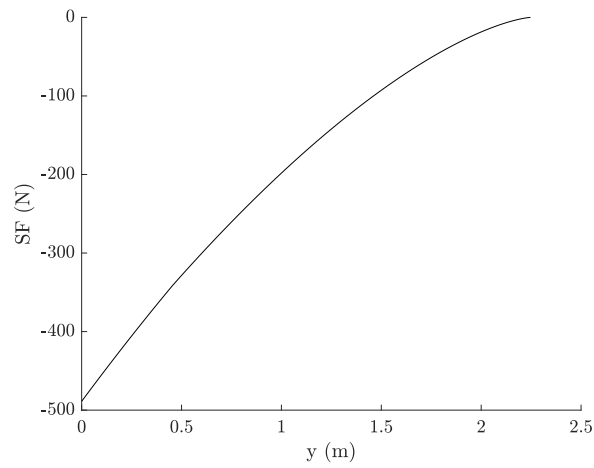


Figure 16: Shear force distribution.

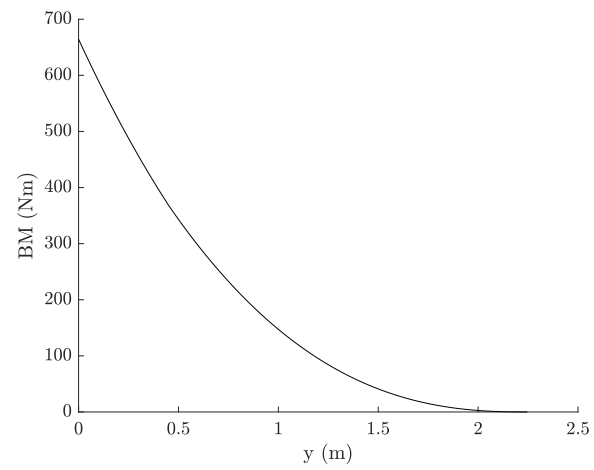


Figure 17: Bending moment distribution.

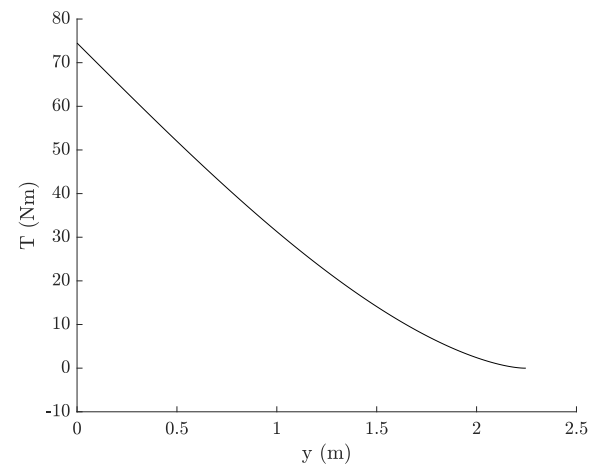


Figure 18: Torque distribution.

C Topology optimisation attempts

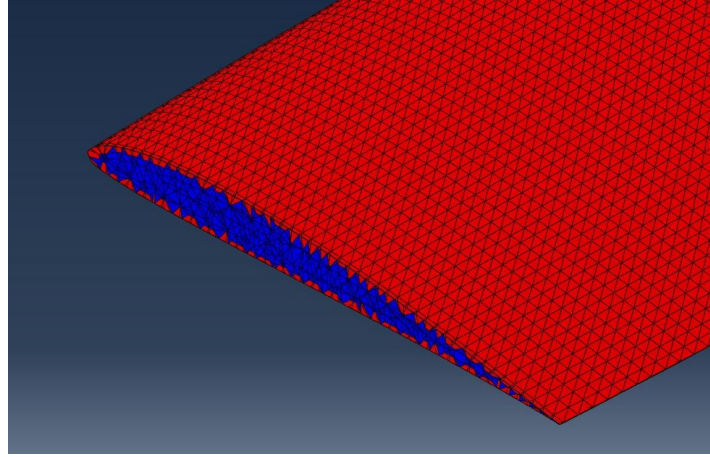


Figure 19: A topology optimisation result with pressure loads applied.

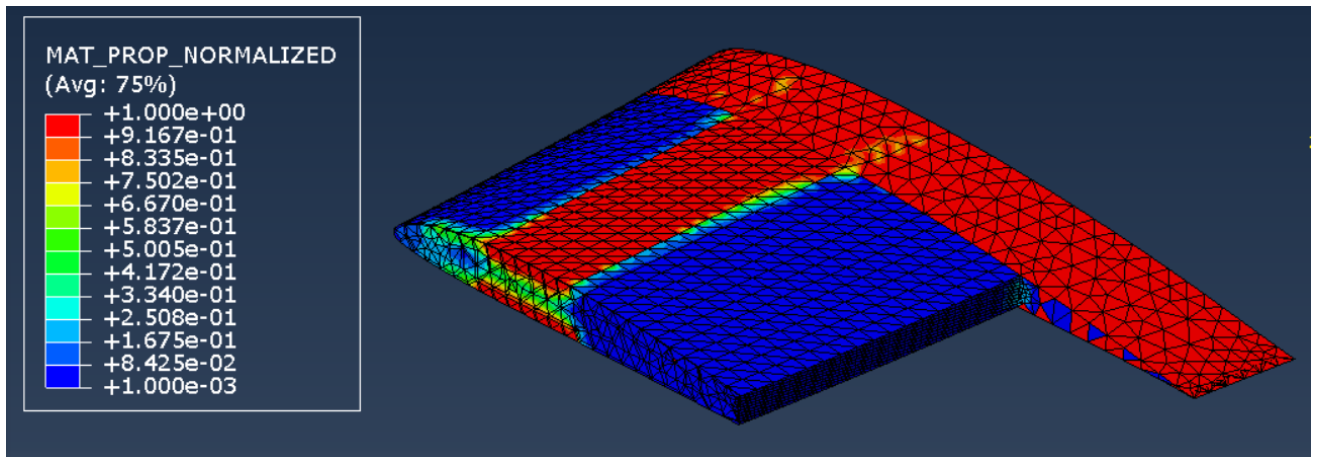


Figure 20: A topology optimisation result with high trailing edge fidelity and low fidelity near the shear centre.

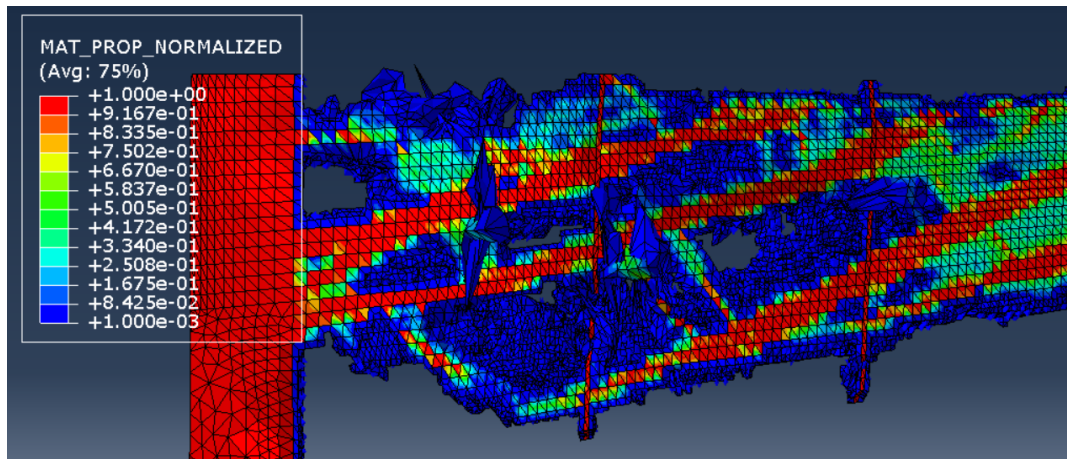


Figure 21: Topology optimisation results for a torsion-only case.

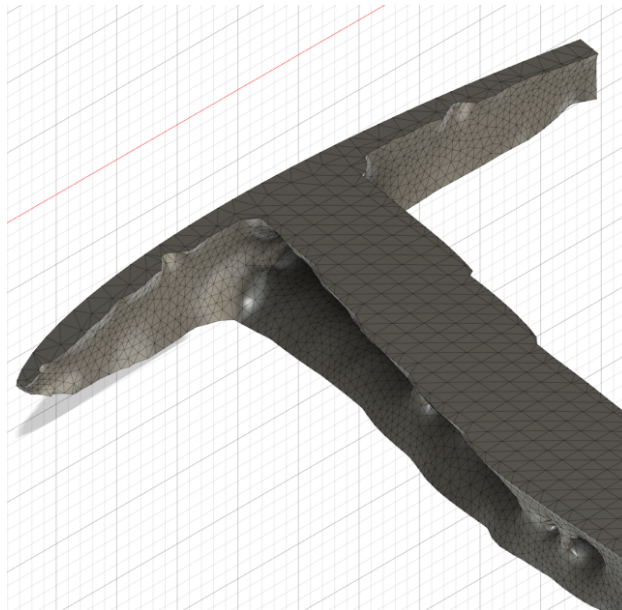


Figure 22: Up-close view of the extracted STL file from topology optimisation near the wing root.

D Buckling

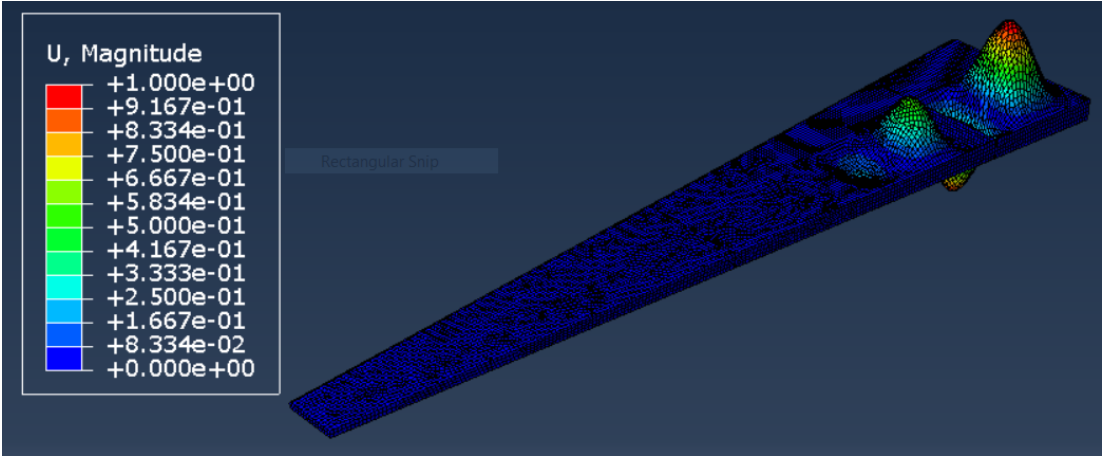


Figure 23: The first buckling mode with buckling of the skin near the root.

E Variable lattice structures in Netfabb

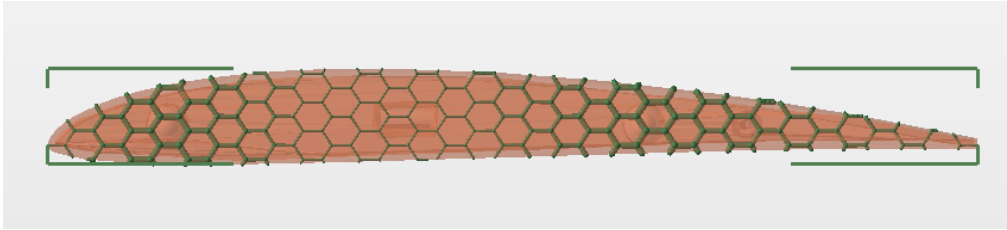


Figure 24: A screenshot taken from Netfabb depicting variable Hexa Grid lattice on a wing cross-section.

F 3D printed parts

The following parts, a wing tip and a rib, were printed with the Department of Aeronautics' at Imperial College London's Ultimaker machines and demonstrated a good fit.



Figure 25: Separated wing tip and rib.



Figure 26: Connected wing tip and rib.